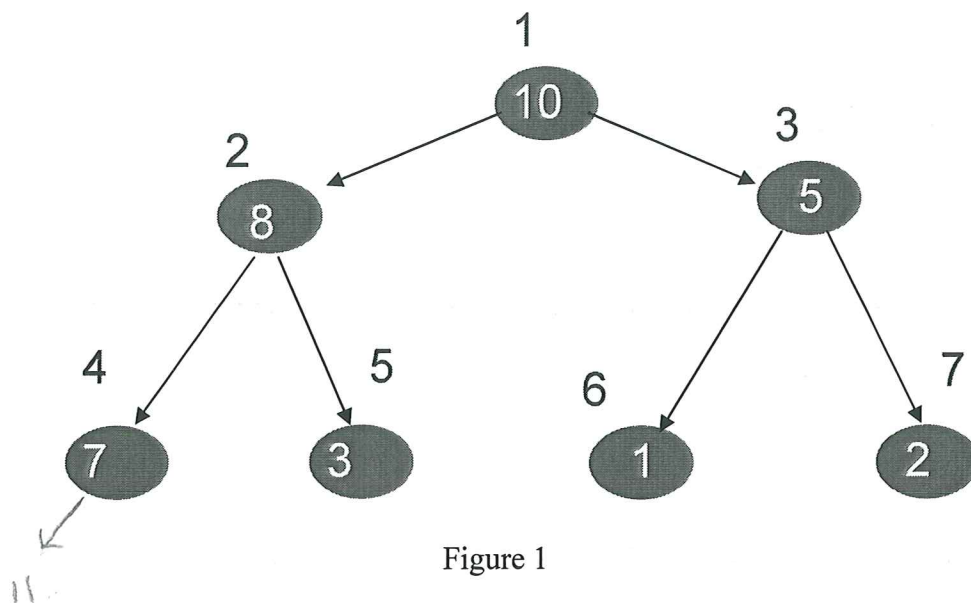


100

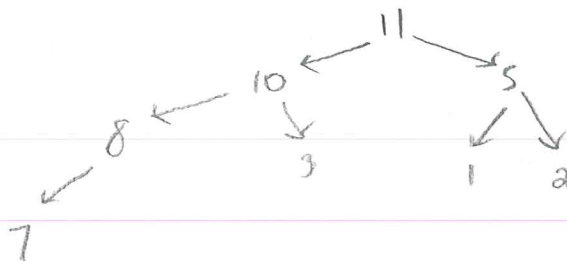
Name Brandon Buchanan

Heap Quiz

Algorithms

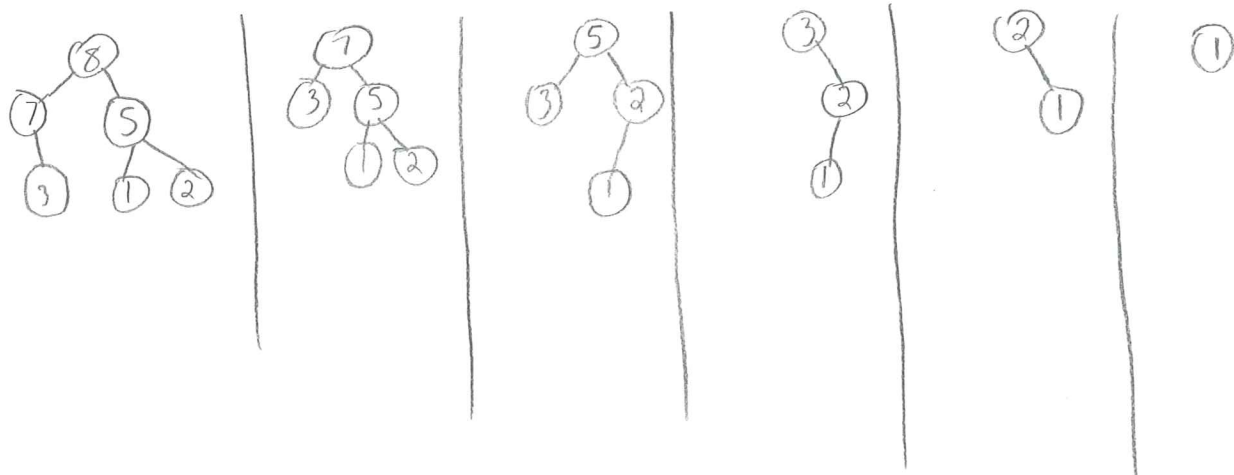


1) Insert the number 11 into the above Figure 1. Find the new heap



2) Start with Figure 1, repeat Extract-Max(H), until there are no nodes left.

10, 8, 7, 5, 3, 2, 1



3) The algorithm to solve 2) is called HeapSort, what is the worst-case running time?

$$n \lg n$$

4) Give an example of a min-heap with 12 nodes?



5) What is the worst case running time for each of the below items in terms of n , the number of nodes?

Assume a min-heap

Make-Heap() creates and returns a new heap containing no elements. 1

Insert(H, x) inserts node x into H . $\lg n$

Minimum(H) returns a pointer to the node in H whose key is minimum 1

Extract-Min(H) deletes the node from H whose key is minimum, returning a pointer to the node $\lg n$

Union(H_1, H_2), aka Merge, creates a new heap that contains all the nodes of heaps H_1 and H_2 . H_1 and H_2 are destroyed by this operation $2n \lg n$

Decrease-Key(H, x, k) assigns to node x within heap h the new key value k , $k \leq$ current key value $\lg n$

Delete(H, x) deletes node x from heap H . $\lg n$